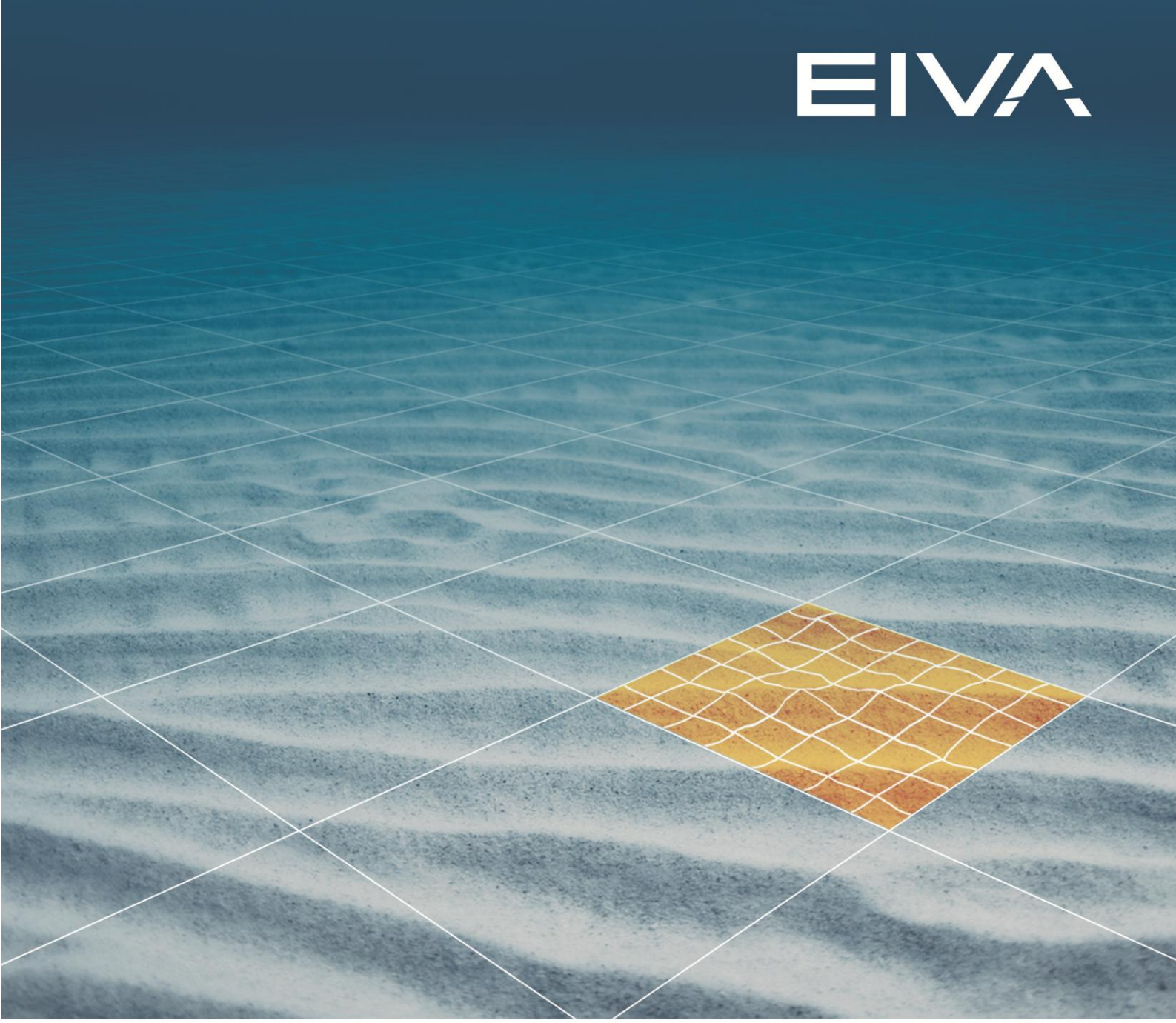

# ROTV VESSEL-AIDED TERRAIN FOLLOW

## USERGUIDE

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# 1 Introduction

Traditionally, EIVA's ROTVs (remotely operated towed vehicles) use a built-in altimeter to follow the seabed and obstacle avoidance. This method, in instances of steep obstacles like shipwrecks and rocky outcrops, potentially avoids obstacles later than ideal.

To further decrease risks of steep obstacles causing sudden movements – or, in the worst-case, collisions – EIVA has developed the **Vessel-Aided Terrain Follow** mode.

With data from a multi-beam echo sounder (MBES) and EIVA NaviSuite Helmsman's online or offline digital terrain model (DTM), a corridor is generated that acts like a false seabed based on the user settings for the Vessel-Aided Terrain Follow. The path smooths steep-sided seabed or obstacle features and is sent as corrections from Helmsman to the ROTV Flight software. This false seabed (the yellow corridor in the image below) reveals obstacles in front of the ROTV earlier than the onboard altimeter does. The onboard altimeter is normally the only sensor used for the following seabed mode, but by allowing the ROTV to look further forward through the **Vessel-Aided Terrain Follow** mode, EIVA strives to achieve safer operations and improved data quality.

The option to use an offline DTM (post-processed) is for situations where there is no hull-mounted MBE present at the towing vehicle to provide an online DTM. The offline DTM should however be used with care, as the seabed could have changed between acquisition and present time, especially if subsea currents are strong, which could have moved underwater objects and sand dunes.

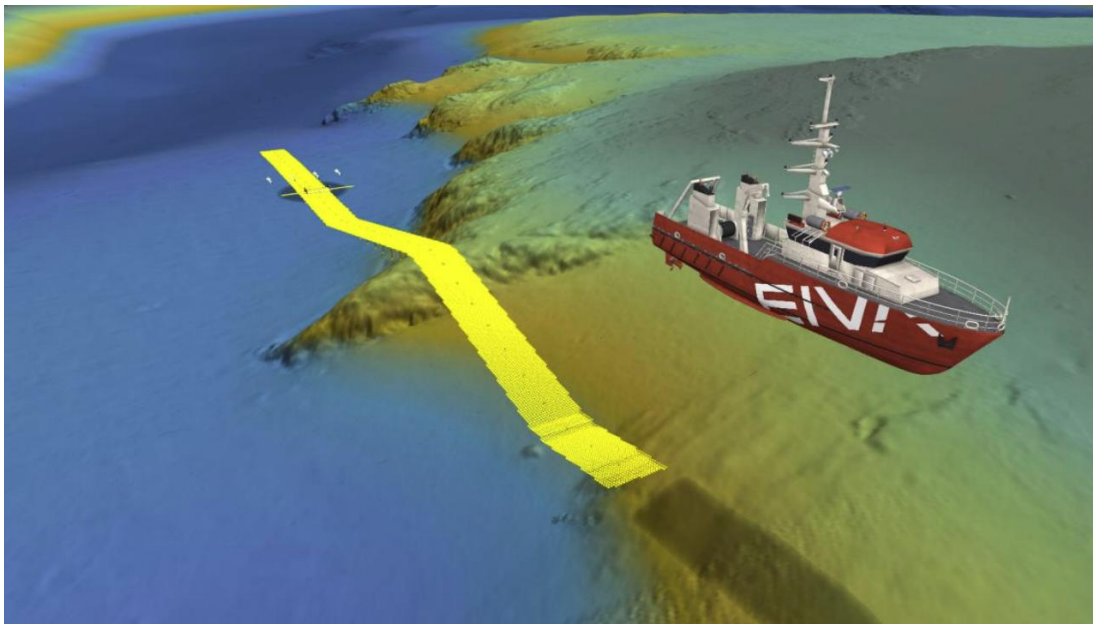

Figure 1 Helmsman, example of Vessel-Aided Terrain Follow mode, showing the corrected route in yellow

The Vessel-Aided Terrain Follow mode can only be used together with EIVA NaviSuite, so no third-party software has this function.

# 2 Preparing the ROTV to receive corrections from Helmsman

Prior to starting the Vessel-Aided Terrain Follow in Helmsman, the ROTV (either ScanFish or ViperFish) needs the function to be enabled on the web interface. Under the **System** tab the **Enable Digitized Seabed** needs to be enabled by setting it to **True**. This option needs to be enabled prior to starting the pre-flight check (). When this option is enabled, it will be remembered, and the setting will remain activated until changed prior to a preflight check in future. When this setting is activated, Flight expects a connection to Helmsman, so it is important to disable it if intending to use the ROTV without the **Vessel-Aided Terrain Follow** mode or with a third-party software.

How to access the web interface and information on different settings, are explained in detail in the ScanFish and ViperFish specific manuals found on EIVA's knowledge base.

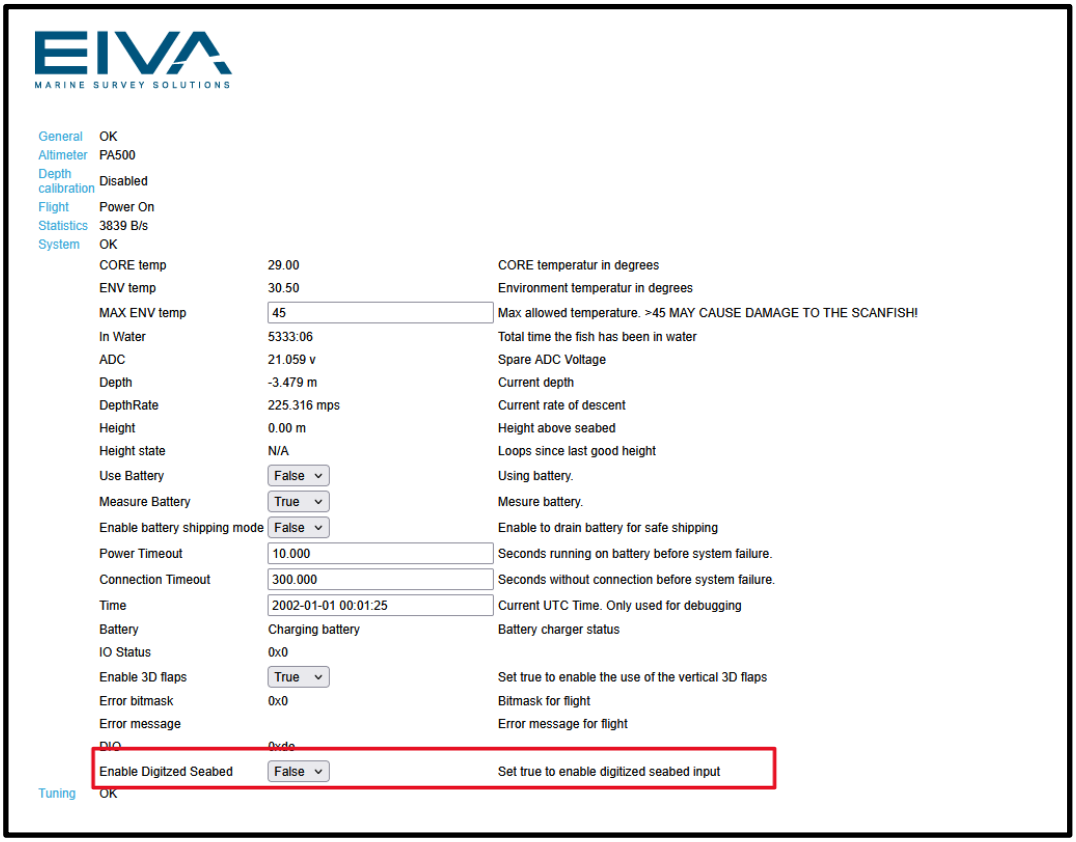


Figure 2 Enable Digitized Seabed on the web interface allowing Vessel-Aided Terrain Follow

When **Enable Digitized Seabed** is set to **True**, and the ROTV is either rebooted or goes through pre-flight check, a new tab will appear in the web interface, called **DTM seabed monitor**. This tab shows all the corrections and settings that are used for Vessel-Aided Terrain Follow mode.

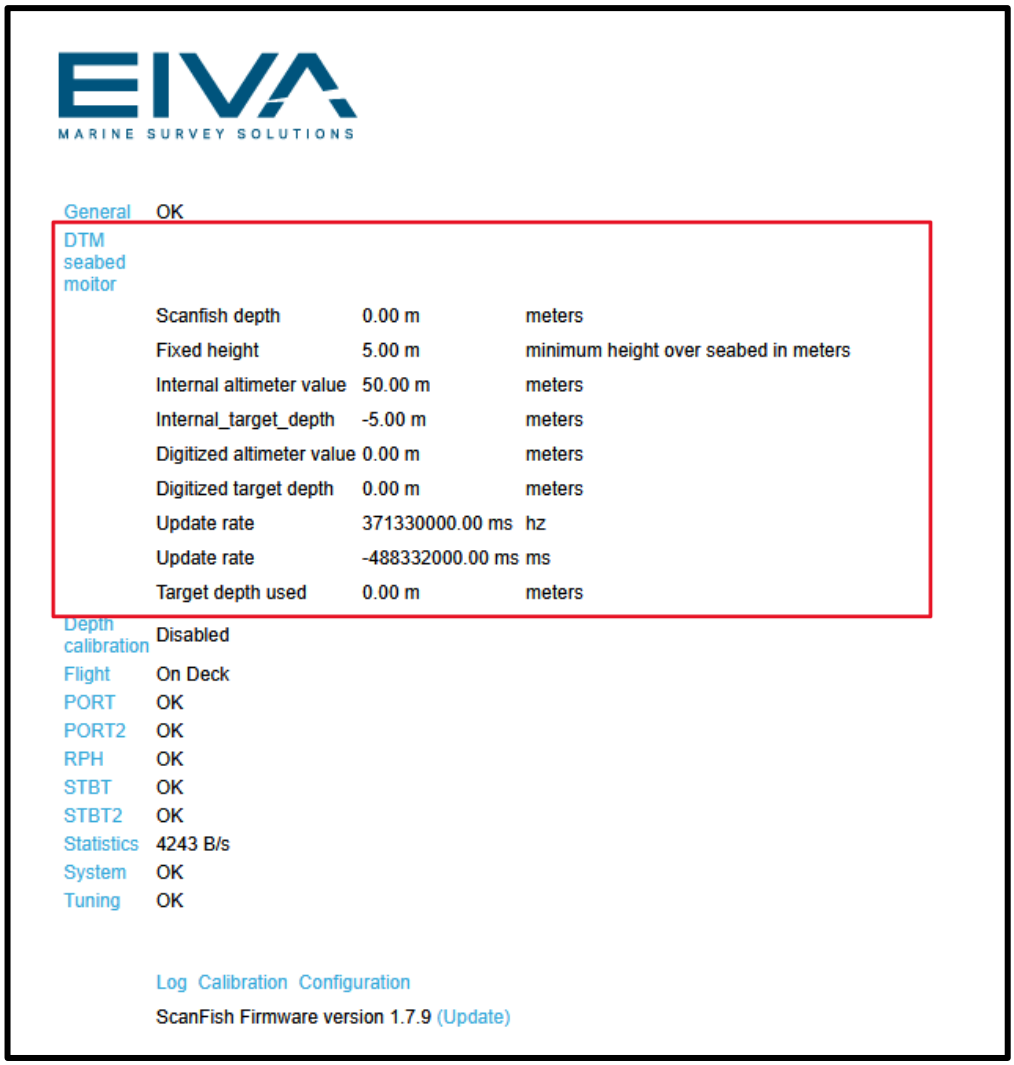


Figure 3 DTM seabed monitor tab in web interface

# 3 Vessel-Aided Terrain Follow mode in Helmsman

To enable the **Vessel-Aided Terrain Follow** module, navigate to the Helmsman **Menu bar** and click **Tools** > **Misc** > **Vessel-Aided Terrain Follow**.

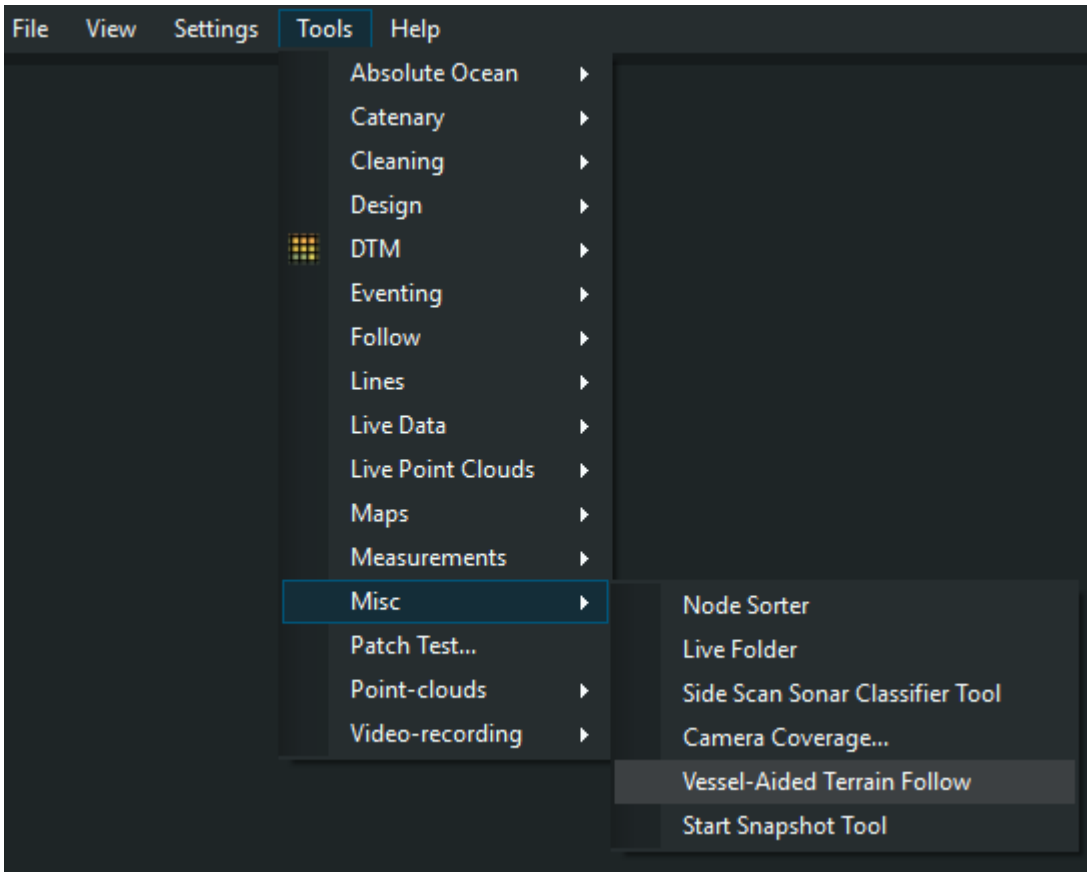


Figure 4 Where to enable Vessel Aided Terrain Follow mode in Helmsman

## 3.1 ROTV connection

After initiating Vessel-Aided Terrain Follow, the ScanFish Connection dialogue box will open.

Figure 5 ScanFish Connection view

- The user should define the ROTV position (**Easting and Northing**) received from the USBL system.
- If the user does not use a USBL system, then instead select the calculated easting and northern positions from the **Generic Cable Counter**. This function is available through setup in **NaviPac configuration**.
- **Target ScanFish** displays the current connection. If the connection is not automatically established, **Target ScanFish** will display **<None>**. When the button is clicked, the view in Figure 6 will appear.
- If the NaviPac computer is on the same network as the ROTV, a connection will automatically be established between NaviPac and the ROTV.

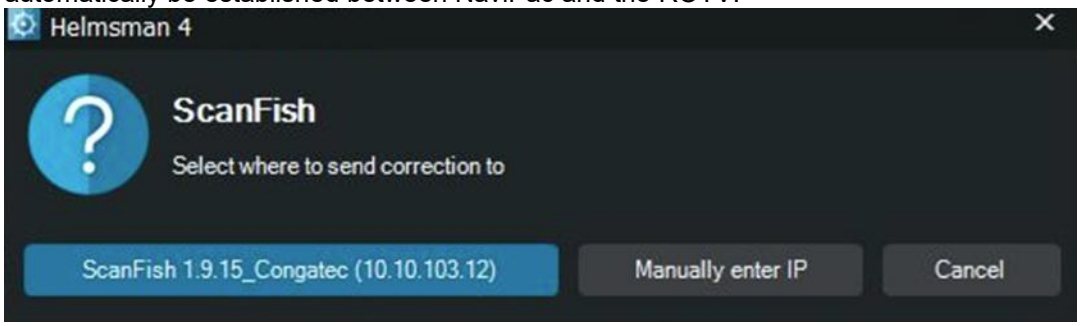


Figure 6 Helmsman select location for correction

Choose between different ROTVs on the network or try to manually connect to a ROTV via the **Manually enter IP** option. When the user establishes a connection to the ROTV manually, you first select the **I/O Type**. Thereafter, type in the IP for the ROTV, which is the standard 10.10.103.XX IP address – where XX is the last two digits on the controller – as shown in Figure 5. The port number is, by default, set to *8844*.

**Note**: If the IP address is not the default one but someone changed it, you can use EIVA program **EivaHWConfig**. The EivaHWConfig program can be found on the Download page: Download Site - EIVA a/s.

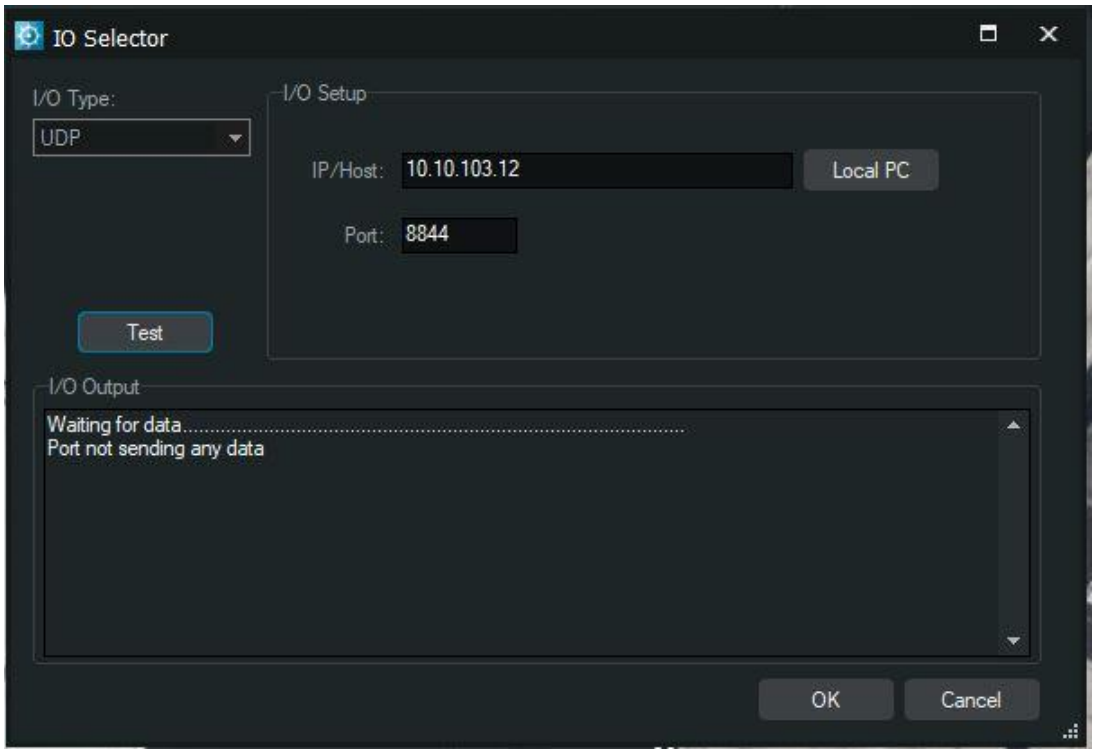


Figure 7 IO Selector view to manually establish connection to the ScanFish

If you press **OK**, the **Vessel-Aided Terrain Follow** view – shown in Figure 8 – will open. This view can be freely docked and will show a **Correction** value in the top. The correction is calculated from the active surface based on the user-defined settings, and the output is sent to ROTV Flight to avoid an obstacle. Details on these settings and how they affect the Vessel-Aided Terrain Follow mode will follow in the next sections.

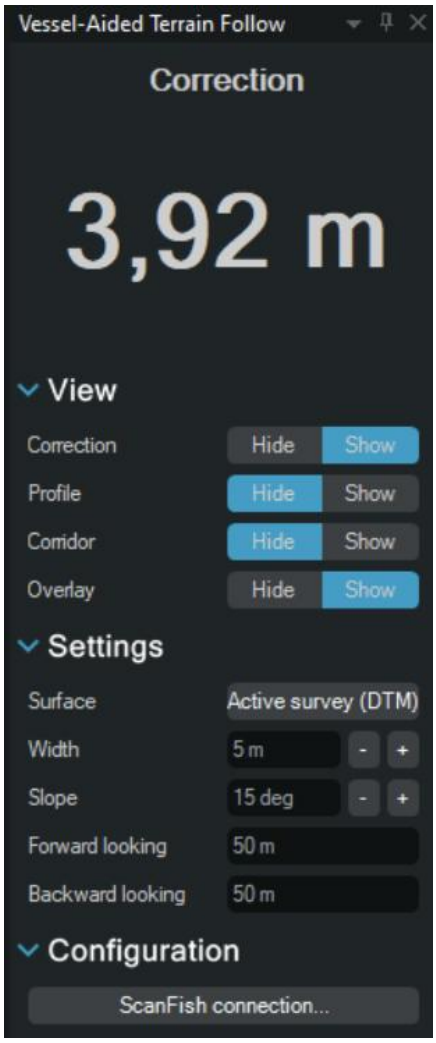

Figure 8 Helmsman layout of the Vessel-Aided Terrain Follow window

## 3.2 Surface method – Corridor or Carpet

There are two surface methods that can be used with Vessel-Aided Terrain Follow. The first one is **Corridor** and is explained in detail during the manual. The second method is **Carpet**, where the user selects only the slope for the ROTV to ascend or descend on.

The **Carpet** method will take the DTM (**Active survey**, **Online DTM** or **Custom**) and overlay a carpet over the whole DTM. This method will impact the accuracy of the existing data, so it is not recommended.

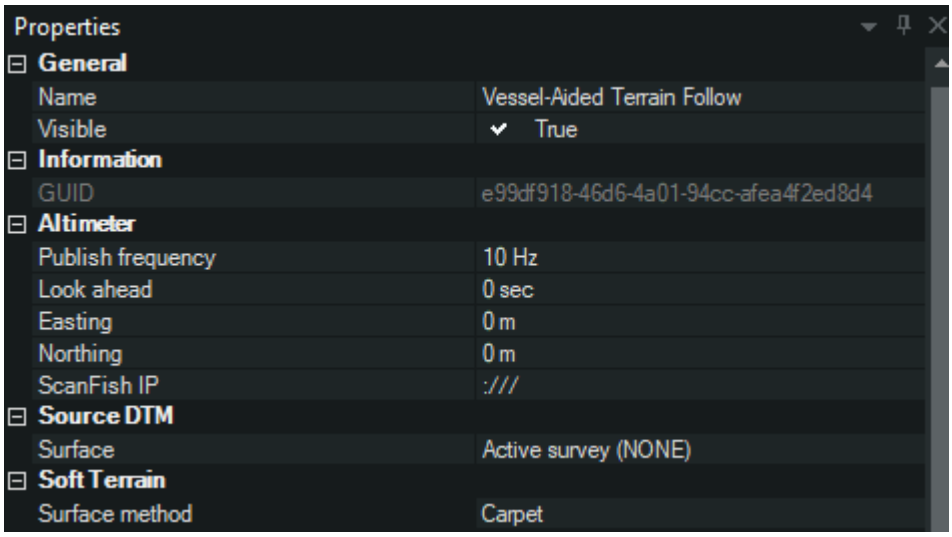


Figure 9 Carpet method is activated in Vessel-Aided Terrain Follow mode Properties window under Soft Terrain

## 3.3 View

Under the **View** section are options to visualise **Correction**, **Profile**, **Corridor** and **Overlay** in the Helmsman **Map View**.

- **Correction** adds two grey ball-shaped objects with a line between them to display the height of the correction.

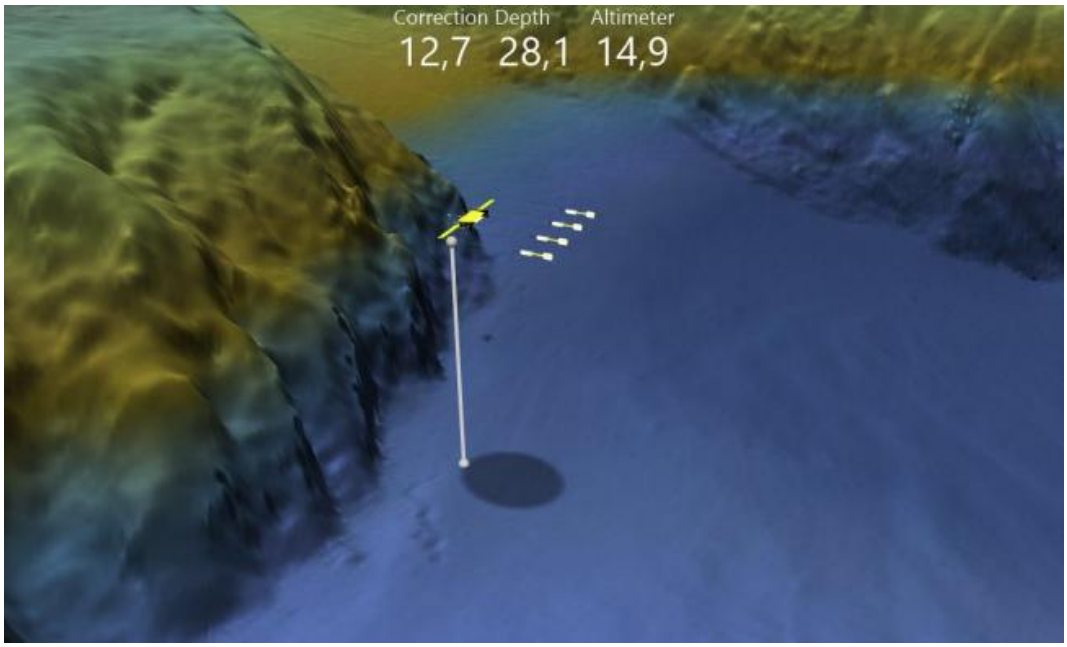


Figure 10 The grey line between the balls shows the correction to ROTV route

- **Profile** displays the height of the correction as pillars. The distance between the pillars is related to the ROTV position update frequency

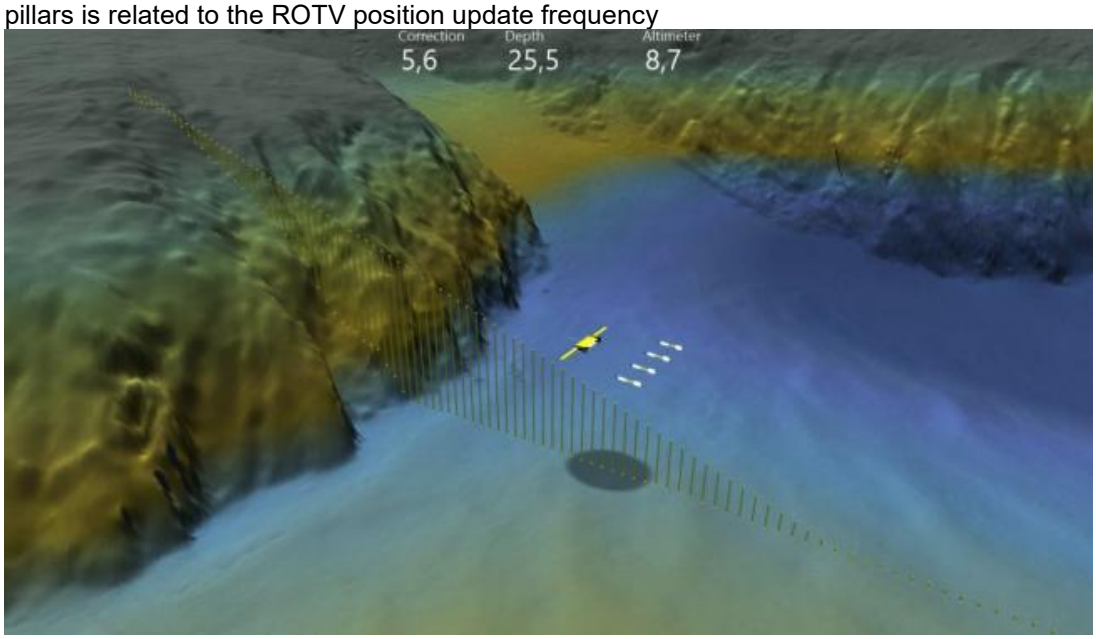


Figure 11 Profile view displays corrections as pillars looking forward

- **Corridor** displays a corridor of the corrected surface which the ROTV keeps a constant altitude relative to (rather than relative to the seabed).

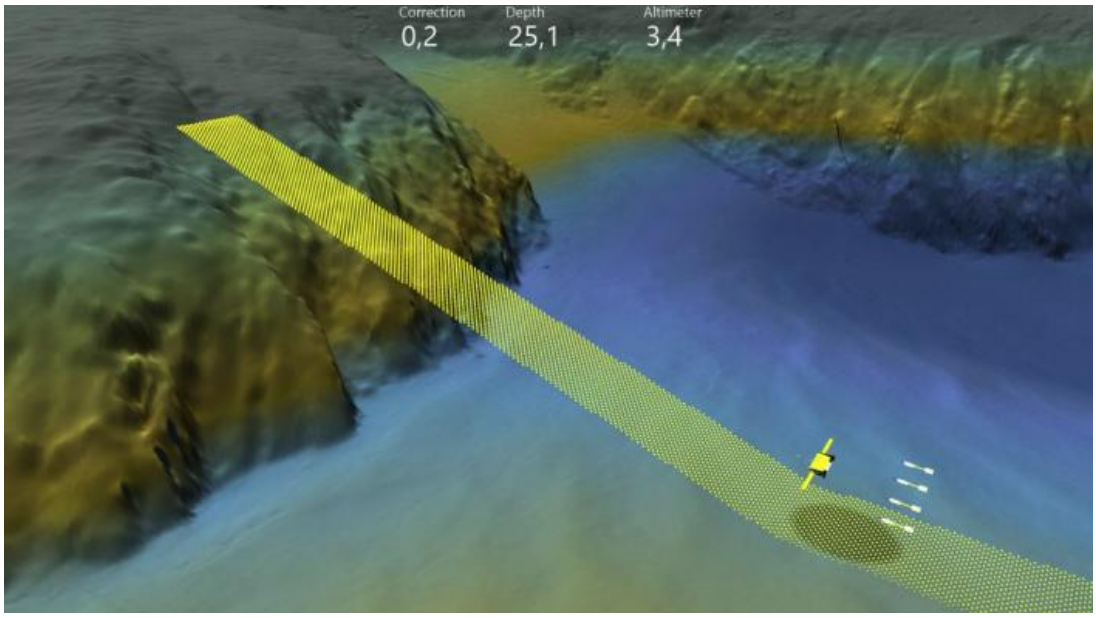


Figure 12 Corrected surface visualised as a corridor

- **Overlay** automatically displays the **Correction** value as an overlay at the top of the **Map View**. This is convenient since, after all settings in the Vessel-Aided Terrain Follow option windows have been configured, you can set the window to **Auto Hide** (by pressing the pin icon in the top of the window),which leaves a cleaner screen and the **Overlay** view enables the user to still monitor the corrections sent to ROTV.

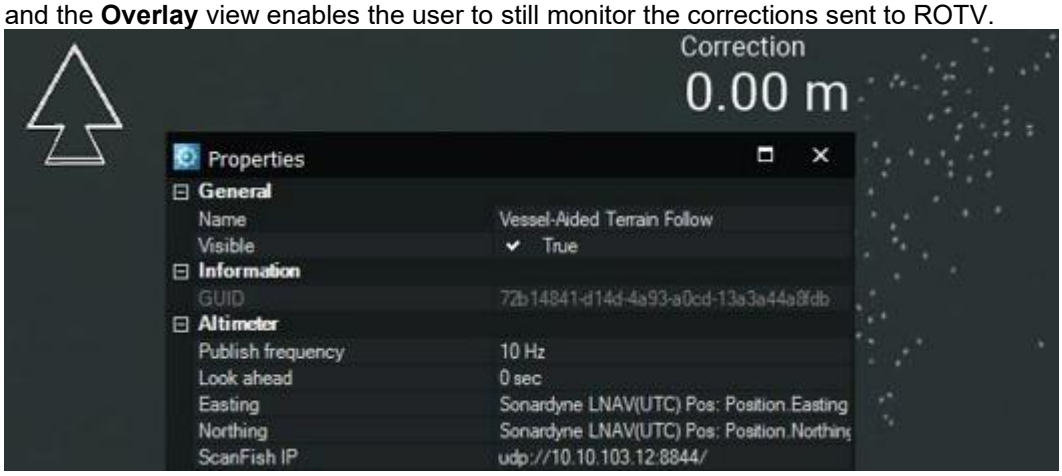


Figure 13 Overlay of the correction shown in top of the map view

## 3.4 Settings

In the **Settings** section, the different settings of Vessel-Aided Terrain Follow mode are described.

**Note**: Changing one of the values will affect the total outcome (seen as the **Correction** value in Figure 8), therefore consider carefully how each component will affect the others.

This is the key to getting safer operation for the equipment and improving the data quality.

### 3.4.1 Surface

There are three options relating to selecting the type of **Surface**, that is the DTM which the Vessel-Aided Terrain Follow mode will calculate the corrections from and that it sends to the ROTV based on Figure 14:

- **Active survey**
- **Online DTM**
- **Custom**

Figure 14 Settings - Surface options

**Active survey** means the currently active DTM – which will be bold in the Helmsman **Project Tree** – is the surface used for calculating corrections.

**Note**: Be careful with this option – when you import a DTM to your project, by default that DTM automatically becomes the active survey surface in the **Project Tree**.

**Online DTM** means the currently built DTM from the MBES via the NaviScan connection is used for corrections. This DTM can have cleaning applied to it in real time to it.

**Note**: Be careful with this option – aggressively applied cleaning could remove obstacles you need to avoid.

**Custom** gives the user the opportunity to use a specific DTM, for instance an old DTM from a pre-survey that is imported to the **Project tree**.

**Note**: To modify a DTM, it must be active in the **Project Tree**, indicated by bold text.

### 3.4.2 Width

It is possible to define the width of the surface considered by Vessel-Aided Terrain Follow when it is calculating corrections. **Width** is the distance (m) to each side of the ROTV position. See Figure 14 under **Settings**.

An example is shown below of the difference between the corrections for a corridor set to a width of 5 m vs 13 m for the same survey line. From the wider 13 m corridor, areas of the rocky outcrop to the right side of the survey line are registered as part of the surface to follow. Consequently, that part of the corrections to the ROTV's route gives it a higher altitude. The narrow 5 m corridor does not detect the outcrop, which is why the smooth DTM has no corrections.

The width should be a compromise between safety of the equipment and potential infill generated by an excessive value. The default value is set to 5 m.

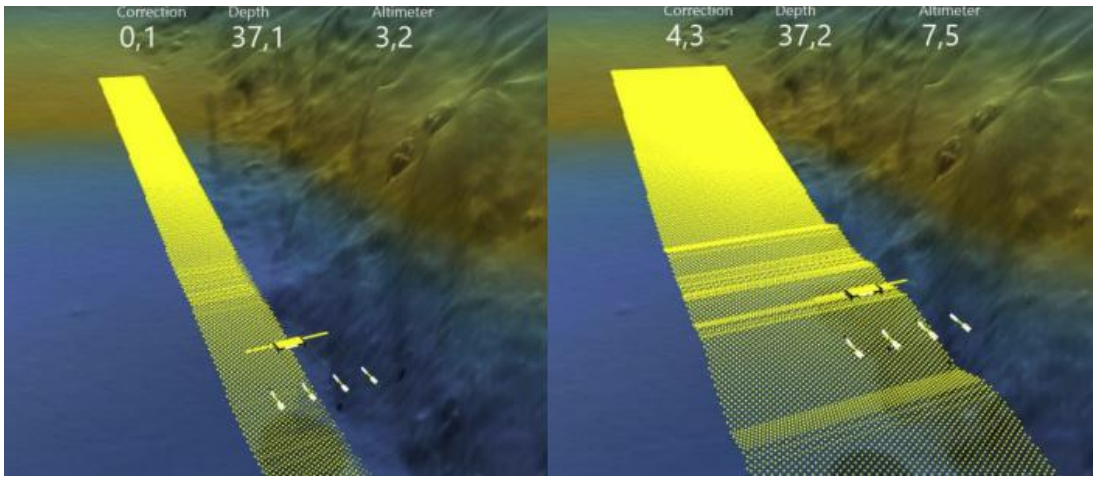


Figure 15 Corridor (smooth DTM) for widths of 5m (left) and 13m (right)

### 3.4.3 Slope

Slope is the maximum angle of the smooth DTM. The ROTV, when trying to avoid an obstacle, will follow a path which does not exceed this angle.

The example in Figure 16 illustrates the difference between a slope angle of 10 and 17 degrees for the same survey line. With a gentler slope, the ROTV ascends earlier, which might be safer, but also leaves a potentially greater gap in the data. By increasing the angle, you could potentially make the inclination too steep for the ROTV to safely ascend and keep the sensors stable during the ascent.

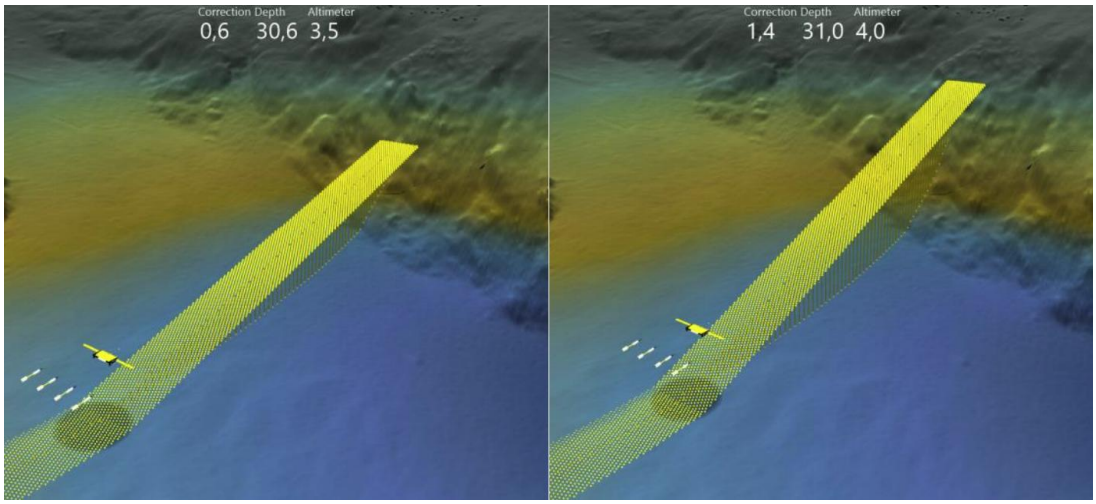


Figure 16 Difference between a corridor with Slope set to 10 degrees (left) versus 17 degrees (right)

There are physical limitations to the ROTV's ability to ascend and descend.

These are mainly a function of the speed through water (STW), cable layback and the setup of sensors on your ROTV. In the Flight software, the default ROTV ascend/descend rate is set to 0.6 m/s.

**Example**: With a ScanFish Katria setup equipped with four soft-towed magnetometers, a STW of 4.0 knots is very common. For this setup, the default ascend/descend rate of 0.6 m/s works well. These values result in an approximate maximum slope angle of 16.25 degrees (see calculation below). The default slope value is set to 15 degrees.

Assuming the user maintains a constant ascend/descend rate, the slope angle setting must be changed accordingly with adjustments to the STW. Use the formula in below to calculate the correct **Slope**.

$$Slope = Tan^{-1}\frac{Genereal\ rate}{\text{STW}}$$

$$General\ rate = 0.6\frac{m}{s} * 0.514 = 1.17\ knt$$

$$Slope = Tan^{-1}\frac{1.17\ knt}{4.0\ \text{knt}} = 16.25°$$

Equation 1 Slope angle calculation

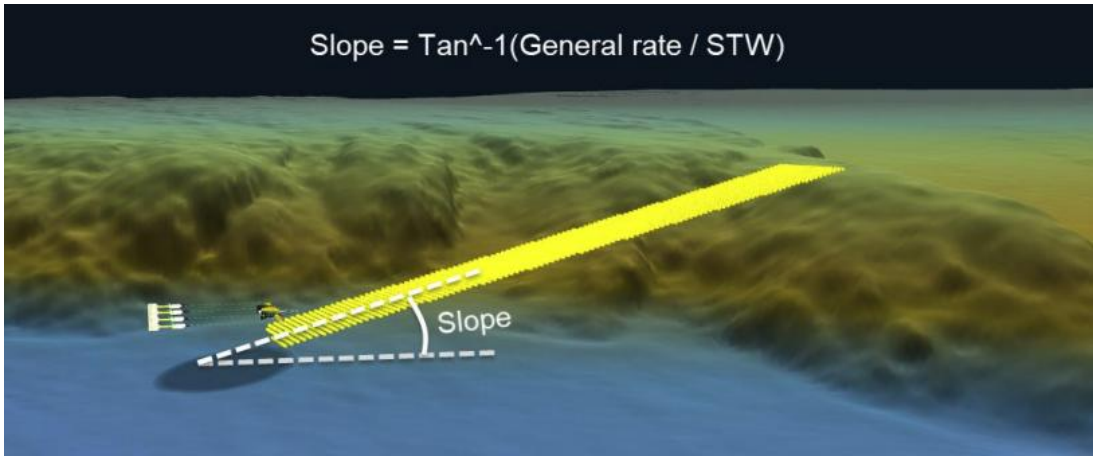


Figure 17 Slope angle and formula to calculate appropriate slope angle

### 3.4.4 Forward looking

This value is mainly used for a controlled ascent of steep-sided obstacles. This defines a distance in metres ahead of the ROTV position. The greater the **Forward looking** distance, the earlier the ROTV will detect a seabed obstacle in front of it and start to compensate for it. This is illustrated in Figure 18, where the ascent happens sooner with a greater value. This function will only account for available data, so if the forward-looking value exceeds the edge of the selected surface, the missing data will be read as *NaN* (not a number), see Figure 19. The default **Forward Looking** is set to 50 metres.

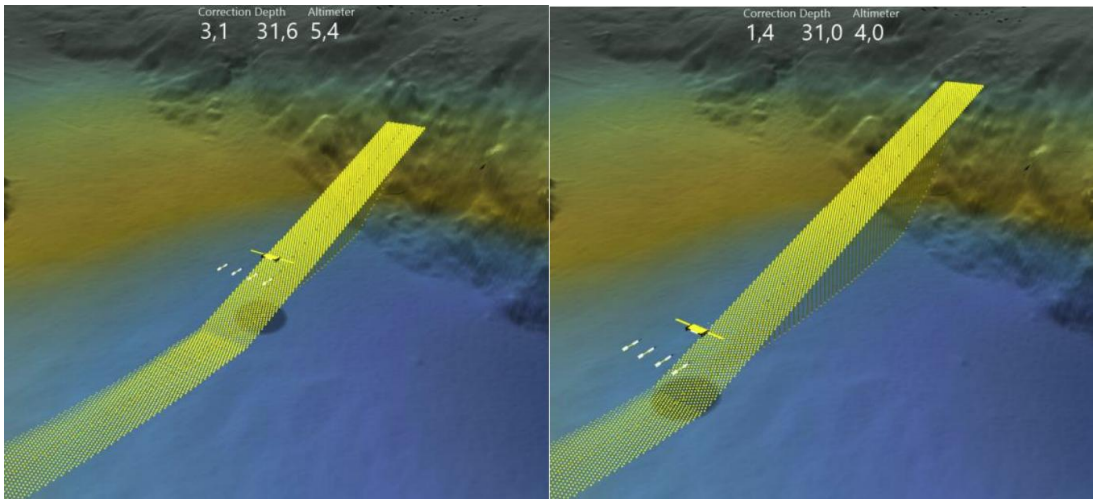


Figure 18 ScanFish ascending toward the right with Forward looking set to 30 m (left) and 50 m (right)

Figure 19 Correction showing NaN when there is no data to calculate the correction from

### 3.4.5 Backward looking

Backward looking is mainly used for a controlled descent at steep-sided drop-offs.

In Figure 20, a large (50 m – left image) and low (10 m – right image) **Backward looking** value is shown for the same survey line. The greater the value used, the more of the selected surface is compensated for, resulting in a steady and more controlled descent (left) compared to the more sudden uncontrolled descent (right).

**Note**: The **Forward-looking** value in both cases is set to 0 m, as this effectively does not affect the descent.

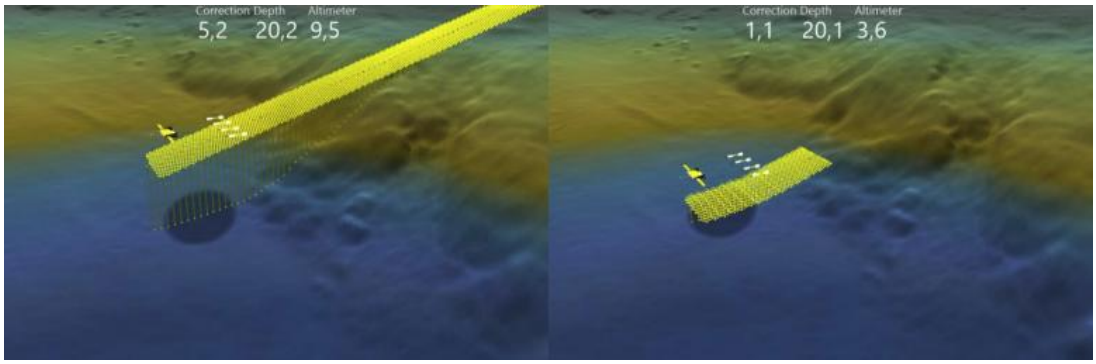

Figure 20 ScanFish descending toward the left – Backward looking set to 50 m (left) and 10 m (right)

## 3.5 Configuration

When you click the **ScanFish connection** button, the **ScanFish Connection** view (shown in Figure 5) will open and the user can set up a new ROTV connection by following the steps in section 2.1 ScanFish connection.

# 4 Flight software

In the Flight software, the **Vessel-Aided Terrain Follow** flight mode has been added to the **Flight Mode** drop-down menu shown when you click on the **Flight Mode** button in the toolbar, as shown in Figure 21.

Figure 21 Flight Mode drop-down with Vessel-Aided Terrain Follow option

The **Vessel-Aided Terrain Follow** mode is similar to the **Follow Seabed** mode in the Flight software. Basically, the ROTV will follow the seabed at a user-defined altitude above the seabed unless corrections are sent across from the NaviPac computer. It uses the same altitude from the **Set Follow Seabed** value in the **Tow** panel (Figure 22), so the ROTV will maintain the desired altitude to the seabed, plus the **Correction** value sent over from Helmsman.

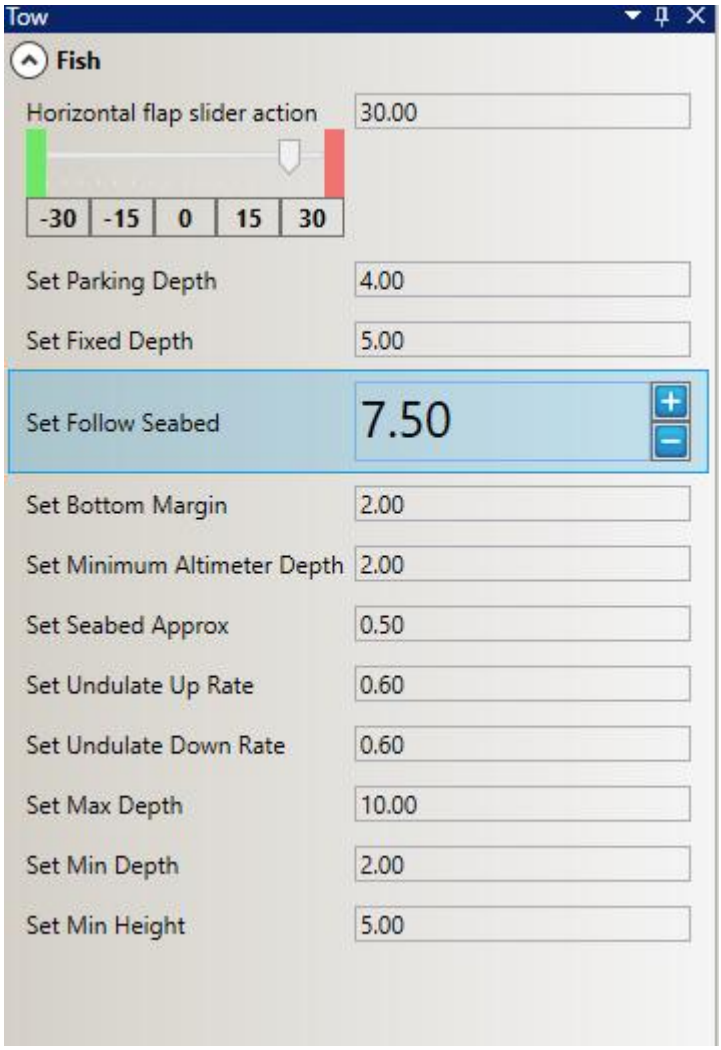


Figure 22 Tow panel: Set Follow Seabed value controlling the altitude during Follow Seabed and Vessel-Aided Terrain Follow modes

As a safety precaution, if the ROTV altimeter's reading is lower than the **Set Bottom Margin** value in the **Tow** panel, the ROTV will make a manoeuvre to quickly get distance to the seabed, the same as when in **Follow Seabed** mode.

## 4.1 NaviSuite Flight when Helmsman Vessel-Aided Terrain follow mode is active

The image below illustrates the behaviour of NaviSuite Flight when the Vessel-Aided Terrain Follow flight mode is active. As described previously, the ROTV follows the seabed using its sensors, then when there is a sharp obstacle registered in Helmsman, it instead follows the correction sent over by Helmsman.

Corrections are sent from Helmsman to the ROTV controller, and all the calculations are done on the controller. Corrections are not transmitted to NaviSuite Flight.

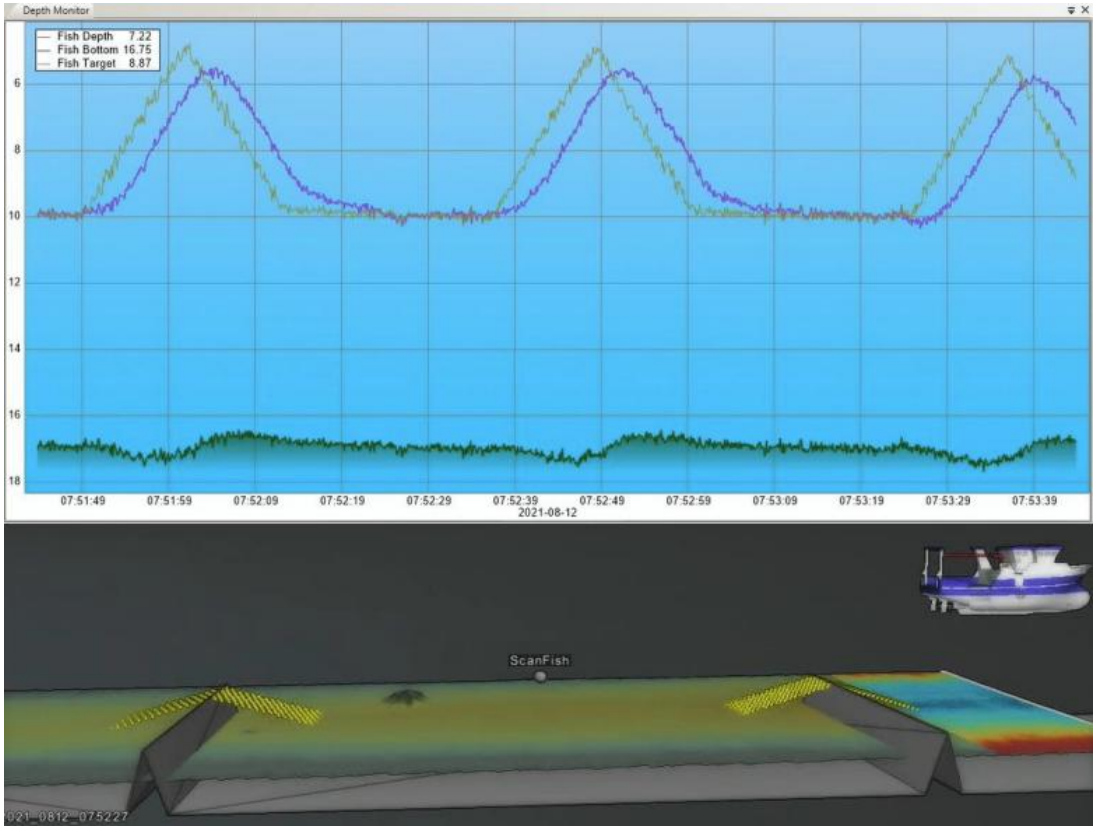


Figure 23 Vessel-Aided Terrain Follow mode – example showing obstacle avoidance in both Flight (top) and Helmsman (bottom)

For this example, a user-generated TIN surface was selected as **Surface** to generate arbitrary seabed obstacles for the ROTV to avoid.

In Helmsman there is an option to **Create bumps model ahead**. This option will generate TIN surface like in Figure 23. This is beneficial if you want to test the Vessel-Aided Terrain Follow mode and how your ROTV behaves to find the best settings. Each ROTV has different behaviours depending on outside factors as currents, different configurations, vessel speed, and more – so it is best to find the applicable settings beforehand.

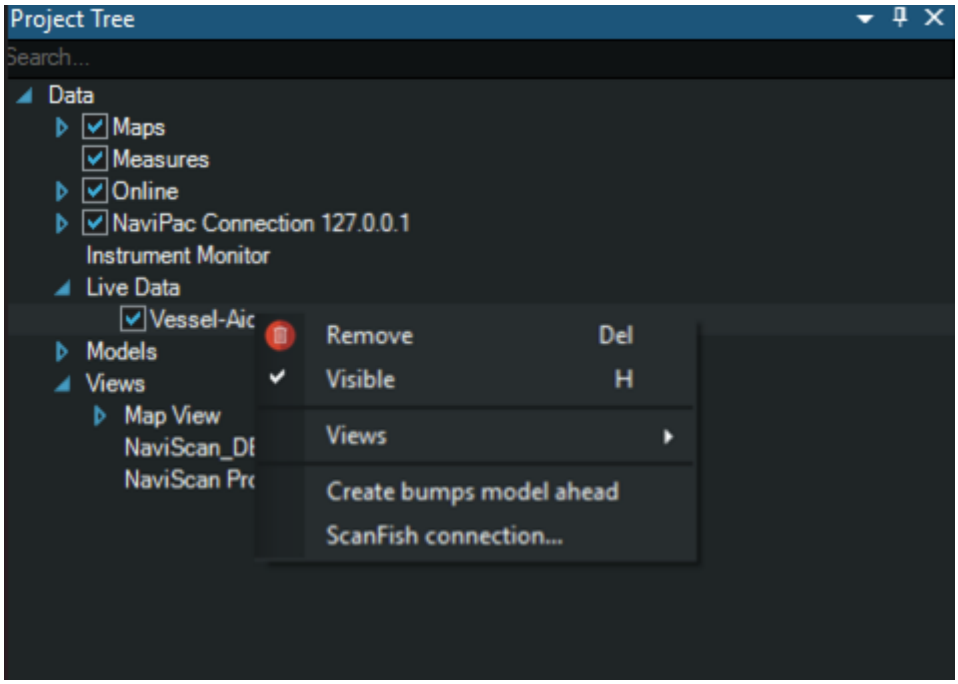


Figure 24 Creating the TIN model for testing Vessel-Aided Terrain Follow

In the Helmsman image, the yellow Corridor of the ROTV's corrected route is shown creating a smooth Surface around the sharp obstacles. In the areas where only the DTM is shown, the Flight software takes over, and the **Fish Target** (orange line) is active in Flight's **Depth Monitor** panel.

When **Vessel-Aided Terrain Follow** mode is active and a sharp obstacle is encountered, the corrections sent from Helmsman take over control instead of Flight's target depth.

The **Fish Bottom** (green line) is the calculated seabed depth generated from the ROTV pressure sensor and altimeter readings. The **Fish Depth** (purple line) is the actual depth of the ROTV.

# 5 Version descriptions

| Version | Date | Author | Description |
|---|---|---|---|
| 1.0 | 08.09.2022 | ASM | Created |
| 1.1 | 18.11.2025 | FSL+ASM | Update |
| | | | |
| | | | |